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DATA ANALYTICS

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ABSTRACT

Hospital readmissions within 30 days of discharge are a major driver of healthcare costs and a focus of financial penalties under the CMS Hospital Readmissions Reduction Program, making early identification of at-risk patients a priority for providers. This project applied the full Data Analytics Lifecycle — Discovery, Data Preparation, Model Planning, Model Building, Communicate Results, and Operationalization — to build a binary classifier predicting 30-day readmission risk in diabetic patients, using the UCI "diabetic_data" dataset of 100,227 hospital encounters across 130 US hospitals.

After removing columns with excessive missingness, engineering three utilisation-based features (total_visits, medication_ratio, lab_ratio), and one-hot encoding categorical variables, the feature space expanded to 2,410 columns. Because only 11.2% of encounters resulted in a 30-day readmission, SMOTE was applied to the training data to correct class imbalance prior to model fitting. Four classifiers — Logistic Regression, Decision Tree, Random Forest, and XGBoost — were trained and evaluated on a held-out test set that preserved the original imbalanced distribution to simulate real-world conditions.

XGBoost achieved the highest ROC-AUC (0.660), narrowly ahead of Logistic Regression (0.630), Random Forest (0.622), and Decision Tree (0.602). However, confusion matrix analysis revealed a critical trade-off: Random Forest correctly identified 518 of 2,237 true readmissions (23.2% recall), more than four times XGBoost's 56 (2.5% recall), despite its lower AUC. This finding demonstrates that ROC-AUC alone is an insufficient selection criterion when Recall — the metric prioritised during the project's Discovery phase — is the clinically meaningful objective. Feature importance analysis identified medication stability (pioglitazone, glyburide, glimepiride, metformin) and prior inpatient utilisation as the strongest predictors of readmission risk.

The project concludes that a deployable readmission-risk model is achievable from routinely collected clinical and administrative data, but that model selection and decision-threshold tuning must be driven by the clinical cost of false negatives rather than by aggregate discrimination metrics alone, in order to translate statistical performance into meaningful reductions in preventable readmissions.

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1. Introduction

Hospital readmissions are widely used as a proxy for the quality of inpatient and post-discharge care. A patient who returns to hospital within 30 days of being discharged may indicate that the original treatment plan was incomplete, that discharge instructions were not followed, or that complications were not anticipated. Beyond the clinical concern, readmissions carry significant financial consequences: under the CMS Hospital Readmissions Reduction Program, hospitals in the United States face reimbursement penalties when their 30-day readmission rates for specific conditions exceed national benchmarks.

Diabetes is a particularly relevant condition for this analysis. Diabetic patients frequently present with comorbidities, complex medication regimens, and variable adherence to outpatient care, all of which contribute to elevated readmission risk. The dataset used in this project — sourced from the UCI Machine Learning Repository — captures ten years of clinical care records for diabetic patients across 130 US hospitals, offering a realistic, large-scale basis for predictive modelling.

1.1 Project Objective

The objective of this project is to build a binary classification model capable of predicting, at the point of discharge, whether a diabetic patient is likely to be readmitted within 30 days. A reliable model allows care teams to prioritise follow-up calls, medication reconciliation, and outpatient scheduling for the patients most likely to benefit from additional intervention.

1.2 Why a Lifecycle Approach

Rather than jumping directly to model training, this project follows the structured Data Analytics Lifecycle — the same framework presented in the accompanying project deck — to ensure that business understanding, data quality, and evaluation rigor are addressed before any model is operationalised. Each subsequent section of this report corresponds to one phase of that lifecycle, mirroring the structure used throughout the project notebook and presentation.

2. Methodology: The Data Analytics Lifecycle

The Data Analytics Lifecycle is a six-phase framework that takes a project from an initial business question through to a deployed, monitored solution. It was used throughout this project, both in the structure of the accompanying presentation and in the organisation of the analysis notebook. The table below summarises how each phase was applied to the readmission prediction problem; the remainder of this report is organised along the same six phases.

Phase	Focus	Key Activities in This Project
1. Discovery	Frame the business problem and success criteria	Define the readmission problem, target population, and the metrics (Recall, Precision, F1, ROC-AUC) that determine success
2. Data Preparation	Clean and structure raw data	Handle missing values, drop low-value columns, engineer new features, encode categorical variables
3. Model Planning	Explore data and select a modelling strategy	Exploratory data analysis, class-balance assessment, correlation analysis, train/test split design
4. Model Building	Train candidate models	Apply SMOTE to correct class imbalance; train Logistic Regression, Decision Tree, Random Forest, and XGBoost
5. Communicate Results	Evaluate and compare models	Classification reports, confusion matrices, ROC curves, feature importance, leaderboard
6. Operationalization	Deploy and monitor	Serialise the best model, verify reload/predict behaviour, define monitoring and refresh plans

Table 2.1 — The six phases of the Data Analytics Lifecycle as applied to this project

This iterative structure is deliberately not a strictly linear pipeline. Insights discovered during Model Planning (Phase 3), for example, frequently send the analyst back to Data Preparation (Phase 2) to engineer additional features — which is exactly what happened in this project with the creation of the `total_visits`, `medication_ratio`, and `lab_ratio` features described in Section 4.

3. Phase 1 — Discovery

3.1 Business Problem

Hospital readmissions increase healthcare costs and can signal inadequate post-discharge care. Because US hospitals are financially penalised for excessive 30-day readmissions under the CMS programme, reducing avoidable readmissions delivers both clinical and financial value.

3.2 Analytics Goal

The Discovery phase translated this business problem into a precise analytics goal: build a binary classifier that predicts whether a diabetic patient will be readmitted within 30 days of discharge, using information available at or before the point of discharge.

3.3 Success Metrics

Before any data was examined, the project defined the metrics by which model performance would be judged. This sequencing matters — it prevents the common pitfall of choosing a metric retroactively to flatter whichever model happens to perform best.

Metric	Definition	Why It Matters Clinically
Recall (Sensitivity)	$TP / (TP + FN)$	Minimises missed high-risk patients — the single most important metric in this clinical context
Precision	$TP / (TP + FP)$	Avoids triggering unnecessary follow-up interventions for low-risk patients
F1 Score	$2 \times (P \times R) / (P + R)$	Balances Recall and Precision into a single comparable number
ROC-AUC	Area under the ROC curve	Measures overall ability to rank readmitted patients above non-readmitted patients, independent of threshold

Table 3.1 — Success metrics defined during the Discovery phase

As flagged in the original notebook, Recall was prioritised as the primary clinical metric: in a readmission-prevention context, the cost of missing a true high-risk patient (a false negative) is considerably higher than the cost of an unnecessary follow-up call to a low-risk patient (a false positive).

4. Phase 2 — Data Preparation

Raw clinical data is rarely analysis-ready, and the `diabetic_data.csv` dataset is no exception. The original 100,227-row, 50-column extract used the string placeholder '?' for missing values — a common quirk of this UCI dataset — which was first converted to a proper NaN representation so that pandas' missing-value tools could be applied consistently.

4.1 Removing Low-Value Columns

Columns missing more than 50% of their values were dropped automatically, since they carry very little reliable predictive signal and can introduce noise during training. Two further identifier columns (`encounter_id` and `patient_nbr`) were removed because they are unique per row and provide no generalisable information. The table below shows the columns most affected by missingness and the action taken for each.

Column	Non-Null Count	Missing %	Action Taken
weight	3,142 / 100,227	96.9%	Dropped (exceeds 50% threshold)
max_glu_serum	5,328 / 100,227	94.7%	Dropped (exceeds 50% threshold)
A1Cresult	16,744 / 100,227	83.3%	Dropped (exceeds 50% threshold)
medical_specialty	51,335 / 100,227	48.8%	Retained — imputed with mode
payer_code	60,182 / 100,227	39.9%	Retained — imputed with mode
race	98,001 / 100,227	2.2%	Retained — imputed with mode

Table 4.1 — Missing-value profile of the most affected columns and the cleaning decision applied

After this cleaning step the dataset shrank from 50 to 45 columns while retaining all 100,227 encounters.

4.2 Creating the Target Variable

The original `readmitted` column contained three categorical values: '<30', '>30', and 'NO'. Because the business objective is specifically about 30-day readmission, this was converted into a clean binary label: 1 for patients readmitted within 30 days, and 0 for all other outcomes (including readmissions that occurred after 30 days). The resulting class distribution is heavily imbalanced — 89,041 encounters (88.8%) were not readmitted within 30 days, while 11,186 (11.2%) were. This imbalance is the single most consequential characteristic of the dataset and shapes nearly every modelling decision described later in this report.

4.3 Handling Remaining Missing Values

With the target variable defined, remaining missing values were imputed using a type-aware strategy: categorical (object-type) columns were filled with their mode (the most frequent category), preserving the dominant class without introducing artificial values; numeric columns were filled with their median, which is robust to the outliers common in healthcare utilisation data (for example, a small number of patients with unusually long hospital stays).

4.4 Feature Engineering

Three new composite features were engineered to express clinically meaningful utilisation signals that are not directly present in any single raw column:

New Feature	Formula	Clinical Meaning
total_visits	outpatient + emergency + inpatient visit counts	Overall recent healthcare utilisation
medication_ratio	$\frac{\text{num_medications}}{(\text{time_in_hospital} + 1)}$	Medication intensity per day of stay
lab_ratio	$\frac{\text{num_lab_procedures}}{(\text{time_in_hospital} + 1)}$	Diagnostic investigation intensity per day of stay

Table 4.2 — Engineered features added during Data Preparation

Adding 1 to the denominator in the ratio features prevents division-by-zero errors for same-day encounters. As Section 8’s feature importance analysis shows, `number_inpatient` — a direct input to `total_visits` — emerged as one of the single strongest predictors in the final model, validating this engineering decision.

4.5 One-Hot Encoding

Machine learning algorithms require numeric inputs, so all remaining categorical columns were one-hot encoded: each unique category was converted into its own binary (0/1) indicator column, with `drop_first=True` used to avoid the “dummy variable trap” of perfect multicollinearity between indicator columns. Because several columns in this dataset — most notably the three diagnosis code columns (`diag_1`, `diag_2`, `diag_3`) and more than twenty individual medication columns — contain dozens to hundreds of distinct categories, this step expanded the feature matrix considerably, from 45 columns to 2,410 columns. Column names were also sanitised to remove characters such as '<', '>', '[' and ']', which would otherwise cause errors in the XGBoost training step performed later.

5. Phase 3 — Model Planning (Exploratory Data Analysis)

Before committing to a modelling strategy, exploratory data analysis (EDA) was used to confirm the scale of the class imbalance and to understand relationships between numeric features.

5.1 Target Class Distribution

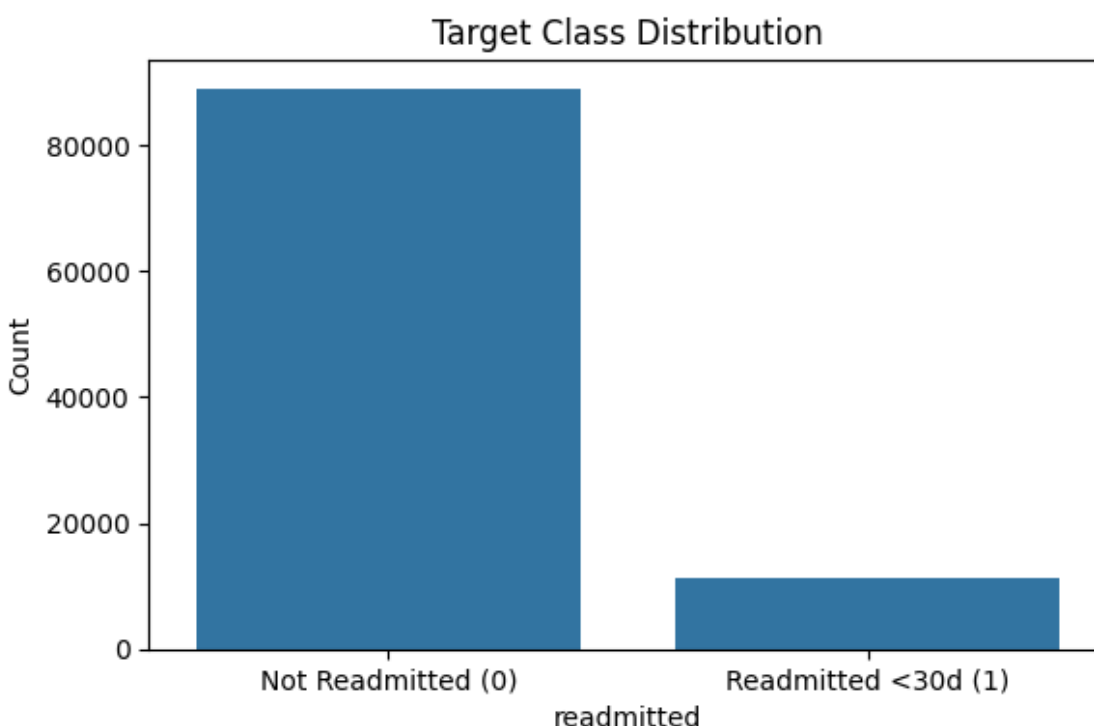


Figure 5.1 — Target class distribution: 89,041 not readmitted vs. 11,186 readmitted within 30 days

The chart confirms what the raw counts in Section 4.2 already suggested: positive cases (readmissions) are outnumbered roughly eight to one. Without correction, a naive model could achieve apparently high accuracy simply by predicting “not readmitted” for every patient — while being clinically useless, since it would never flag a single at-risk patient. This observation directly motivates the use of SMOTE in Section 6.2.

5.2 Correlation Analysis

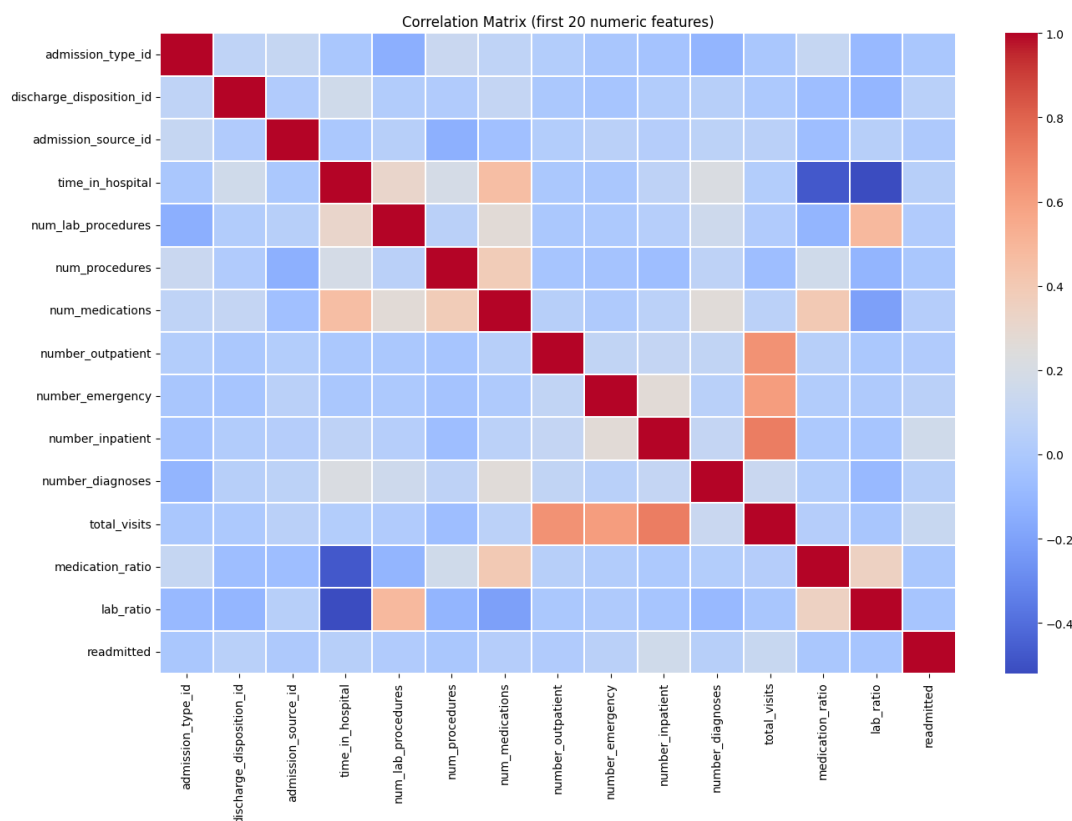


Figure 5.2 — Correlation matrix of the first 20 numeric features, including the engineered `total_visits`, `medication_ratio` and `lab_ratio`

No single numeric feature shows a strong linear correlation with the `readmitted` target in isolation — expected for a clinical outcome that depends on complex, non-linear interactions between medication history, prior utilisation, and diagnosis codes. The heatmap does, however, reveal an intuitive relationship between `number_inpatient`, `number_emergency`, `number_outpatient` and the engineered.

5.3 Train / Test Split Design

A stratified 80/20 train-test split was used, with `stratify=y` ensuring that both partitions preserve the original 11.2% readmission rate. Without stratification, random sampling could by chance concentrate positive cases disproportionately in one split, distorting both training and evaluation.

Split	Rows	Positive Rate (Readmitted)
Training set	80,181 (80%)	11.16%
Test set	20,046 (20%)	11.16%

Table 5.1 — Stratified train/test split summary

6. Phase 4 — Model Building

6.1 Why SMOTE Was Required

Training directly on the imbalanced 88.8% / 11.2% split risks producing a model that is statistically accurate but clinically blind to the minority (readmitted) class. To address this, the Synthetic Minority Over-sampling Technique (SMOTE) was applied — strictly to the training set only — to generate synthetic minority-class examples by interpolating between existing readmitted patients in feature space, rather than simply duplicating them. The test set was deliberately left untouched and imbalanced, so that final evaluation reflects real-world conditions rather than an artificially balanced one.

After SMOTE, the training set grew from its original imbalanced composition to 71,232 negative and 71,232 positive examples — a perfectly balanced training distribution — while the 20,046-row test set retained its natural 11.16% positive rate.

6.2 Candidate Models

Four classifiers were trained on the SMOTE-balanced training data, deliberately spanning a progression from simple and interpretable to complex and high-capacity:

Model	Complexity	Interpretability	Notes
Logistic Regression	Low	High	Wrapped in a Pipeline with StandardScaler to ensure solver convergence
Decision Tree	Medium	High	max_depth = 8 to limit overfitting
Random Forest	High	Medium	300 trees, max_depth = 10, trained in parallel (bagging)
XGBoost	High	Low	Gradient-boosted trees built sequentially (boosting)

Table 6.1 — The four candidate models trained in this project

Tree-based models (Decision Tree, Random Forest, XGBoost) operate on value thresholds rather than distances, so they do not require feature scaling. Logistic Regression, by contrast, relies on a gradient-based solver that converges far more reliably — and considerably faster — once every feature has been standardised to zero mean and unit variance, which is why it alone was wrapped inside a scaling pipeline.

7. Phase 5 — Communicate Results: Model Evaluation

All four models were evaluated on the same held-out test set, which preserved the original 11.16% readmission rate in order to simulate real-world deployment conditions rather than the artificially balanced training distribution.

7.1 Classification Reports

Precision, recall and F1-score were calculated separately for each class. Across all four models, performance on the majority “Not Readmitted” class is strong, while performance on the minority “Readmitted <30d” class — the class that matters clinically — is substantially weaker, a direct consequence of the underlying 8:1 class imbalance reasserting itself once SMOTE’s synthetic balance is removed at evaluation time.

Model	Class	Precision	Recall	F1-score	Support
Logistic Regression	Not Readmitted	0.89	1.00	0.94	17,809
	Readmitted <30d	0.49	0.03	0.06	2,237
Decision Tree	Not Readmitted	0.90	0.97	0.93	17,809
	Readmitted <30d	0.27	0.10	0.15	2,237
Random Forest	Not Readmitted	0.90	0.86	0.88	17,809
	Readmitted <30d	0.18	0.23	0.20	2,237
XGBoost	Not Readmitted	0.89	1.00	0.94	17,809
	Readmitted <30d	0.46	0.03	0.05	2,237

Table 7.1 — Per-class precision, recall, F1-score and support for all four models on the test set

7.2 Confusion Matrices

Confusion matrices break the aggregate metrics above into the four underlying outcome counts — true negatives, false positives, false negatives and true positives — and are essential reading for this project because the original Discovery phase (Section 3.3) identified Recall, not ROC-AUC or accuracy, as the primary success metric.

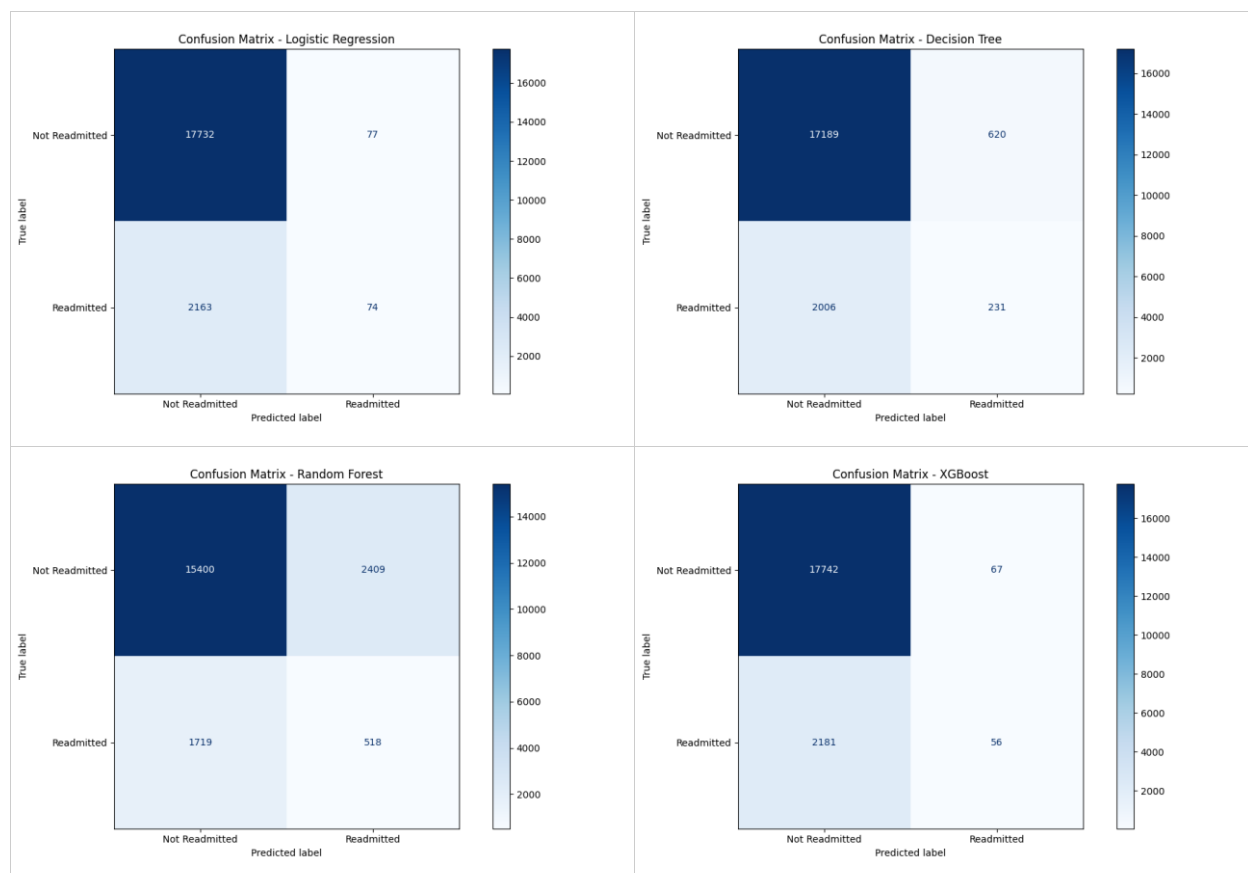


Figure 7.1 — Confusion matrices for Logistic Regression, Decision Tree, Random Forest and XGBoost (clockwise from top-left)

Model	True Negative	False Positive	False Negative	True Positive	Recall (Positive Class)
Logistic Regression	17,732	77	2,163	74	3.3%
Decision Tree	17,189	620	2,006	231	10.3%
Random Forest	15,400	2,409	1,719	518	23.2%
XGBoost	17,742	67	2,181	56	2.5%

Table 7.2 — Confusion matrix breakdown and recall on the positive (readmitted) class, out of 2,237 true readmissions

This table contains the most important finding in the entire evaluation phase: Random Forest, despite having the second-lowest ROC-AUC of the four models, correctly identifies more than four times as many true readmissions (518) as the top-ranked XGBoost model (56). XGBoost and Logistic Regression both achieve high accuracy and respectable AUC largely by predicting “Not Readmitted” for almost every

patient — which satisfies the accuracy metric but defeats the clinical purpose defined in Phase 1. This trade-off is discussed further in Section 10.

7.3 Model Leaderboard (ROC-AUC)

Ranking models purely by ROC-AUC — the discrimination metric most commonly reported in machine learning benchmarks — places XGBoost first, narrowly ahead of Logistic Regression, Random Forest, and Decision Tree.

Rank	Model	ROC-AUC	Test Accuracy
1	XGBoost	0.660	0.89
2	Logistic Regression	0.630	0.89
3	Random Forest	0.622	0.79
4	Decision Tree	0.602	0.87

Table 7.3 — Final model leaderboard ranked by ROC-AUC on the test set

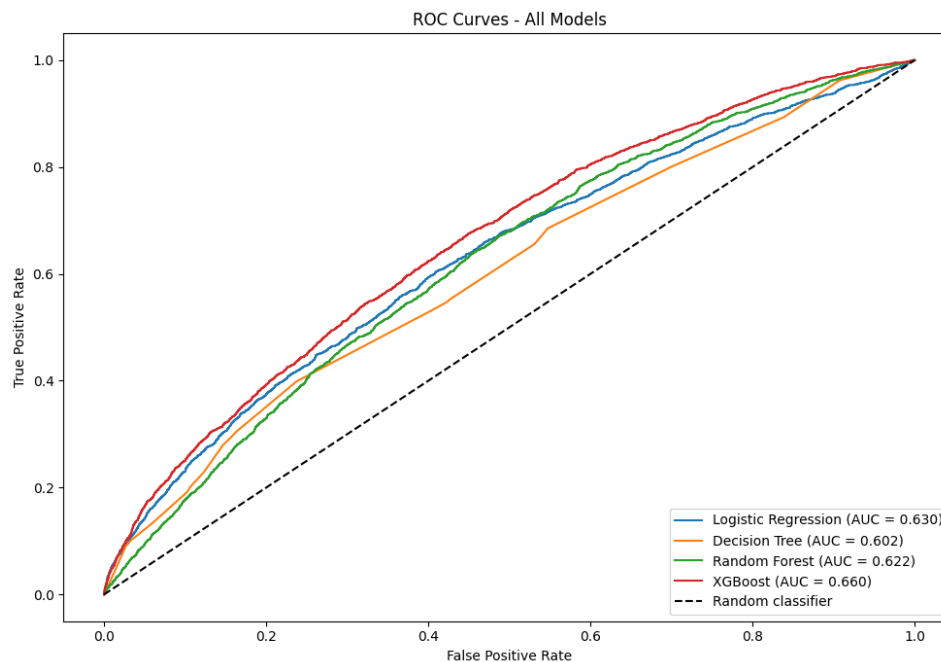


Figure 7.2 — ROC curves for all four models against the random-classifier baseline

8. Feature Importance & Clinical Interpretation

Because XGBoost achieved the highest ROC-AUC, it was selected by the notebook's automated logic as the “best model,” and its feature importances were extracted to understand which variables drive its predictions.

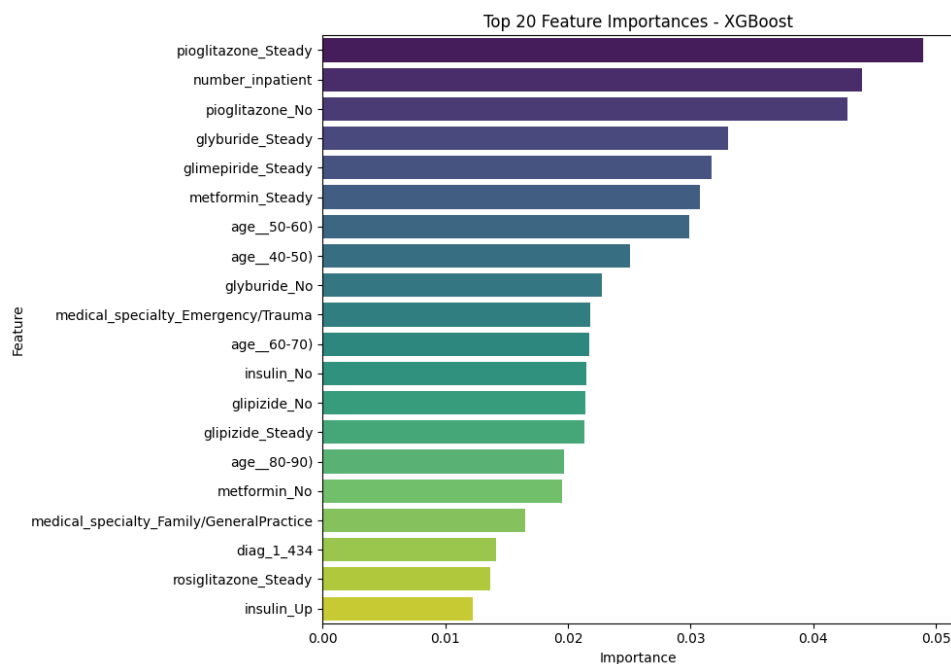


Figure 8.1 — Top 20 feature importances from the XGBoost model

Rank	Feature	Importance
1	pioglitazone_Steady	0.0490
2	number_inpatient	0.0439
3	pioglitazone_No	0.0428
4	glyburide_Steady	0.0330
5	glimepiride_Steady	0.0317
6	metformin_Steady	0.0307
7	age__50-60)	0.0299
8	age__40-50)	0.0250
9	glyburide_No	0.0227

Rank	Feature	Importance
10	medical_specialty_Emergency/Trauma	0.0218

Table 8.1 — Top 10 most important features driving XGBoost predictions

Three patterns stand out. First, diabetes medication status — specifically whether pioglitazone, glyburide, glimepiride and metformin dosages were held “steady” during the encounter — dominates the top of the list, suggesting that medication stability (or instability) around the time of admission carries strong predictive signal. Second, number_inpatient (prior inpatient visits) is the single strongest non-medication feature, reinforcing the value of the total_visits engineering decision made in Section 4.4: patients with a history of frequent inpatient stays are measurably more likely to return. Third, age bands in the 40–60 range feature prominently, consistent with clinical literature suggesting that mid-life diabetic patients with comorbidities are at elevated readmission risk.

9. Phase 6 — Operationalization

The final lifecycle phase moves the analysis from a notebook experiment to an artefact that a real system can consume. The best-performing model (selected by ROC-AUC, as the highest-ranked option from Section 7.3) was serialised to disk using joblib, which is preferred over Python's built-in pickle module for its faster, more memory-efficient handling of the large NumPy arrays embedded inside trained scikit-learn and XGBoost models.

9.1 Deployment Verification

Saving a model is not sufficient on its own — a model that fails to reload correctly is worthless in production. The notebook therefore reloaded the serialised model from disk and ran a sanity-check prediction on five held-out test rows, confirming that the model could be loaded and used for inference without retraining. This reload-and-verify step is a minimum bar for any model intended for integration into a hospital decision-support system.

9.2 Integration Considerations

In a production setting, this model would typically be invoked at or shortly before the point of discharge, taking the same engineered feature set (medication status, prior utilisation counts, demographic and diagnosis information) computed from the electronic health record. The model's output — a readmission-risk probability — would feed into a clinical workflow, such as flagging a patient for a structured follow-up call, a medication reconciliation review, or an outpatient appointment within seven days of discharge.

Given the recall/AUC trade-off identified in Section 7.2, any operational deployment should treat the model's default 0.5 probability threshold as a starting point rather than a final answer; Section 10 expands on how threshold selection should be revisited before go-live.

10. Key Findings & Business Impact

10.1 Key Findings

- Replacing the '?' placeholder with proper NaN values and removing columns with more than 50% missingness materially improved data quality before any modelling began.
- The engineered total_visits, medication_ratio and lab_ratio features captured clinically meaningful utilisation signal not present in any single raw column, and number_inpatient (a component of total_visits) ranked among the top predictive features.
- SMOTE successfully corrected the 88.8%/11.2% class imbalance in the training data, allowing all four models to learn patterns associated with the minority (readmitted) class rather than defaulting to the majority class.
- All four models were evaluated on an identical, untouched, imbalanced test set, ensuring a fair, apples-to-apples comparison that reflects real-world deployment conditions.
- Ranked purely by ROC-AUC, XGBoost is the top performer (0.660); however, ranked by Recall on the clinically critical positive class — the metric prioritised during Discovery — Random Forest is the clear leader, catching 23.2% of true readmissions versus XGBoost's 2.5%.
- Medication stability indicators (pioglitazone, glyburide, glimepiride, metformin) and prior inpatient utilisation are the dominant drivers of predicted readmission risk.

10.2 Business Impact

Healthcare providers can use a model of this kind to flag high-risk patients at the point of discharge and prioritise follow-up care — potentially reducing 30-day readmission rates, the associated CMS financial penalties, and the clinical harm associated with preventable readmissions. The magnitude of that impact, however, depends entirely on which model and which decision threshold are chosen for deployment: a model tuned purely for ROC-AUC, as this project's automated “best model” selection logic does by default, would in practice miss the overwhelming majority of patients who go on to be readmitted, undermining the very business objective that motivated the project. This makes the recommendations in Section 11 essential reading before any production decision is made.

11. Limitations & Recommendations

11.1 Limitations

- Overall predictive performance is modest (ROC-AUC 0.60–0.66 across all models), reflecting the genuinely difficult, multi-causal nature of hospital readmission and the limits of what is captured in administrative and medication data alone.
- The automated “best model” selection in this project used ROC-AUC as the sole ranking criterion, which — as shown in Section 7.2 — does not align with the Recall-first success criteria defined during Discovery (Section 3.3).
- All models were evaluated at the default 0.5 probability threshold; no threshold tuning, cost-sensitive learning, or hyperparameter search was performed for any of the four algorithms.
- One-hot encoding diagnosis codes and 20+ medication columns produced a very high-dimensional feature space (2,410 columns) relative to the number of positive examples, which can make tree-based models prone to latching onto sparse, high-cardinality categories.
- The dataset spans encounters across ten years and 130 hospitals; the analysis does not adjust for hospital-level or temporal variation in care practices, which may confound feature relationships.

11.2 Recommendations

1. Re-rank models using Recall or F1 on the positive class — not ROC-AUC alone — as the primary selection criterion, consistent with the success metrics defined in Phase 1.
2. Tune the classification decision threshold (rather than using the default 0.5) for the selected model, using the precision-recall trade-off curve to pick a threshold aligned with the hospital’s operational capacity for follow-up interventions.
3. Apply class-weighting or cost-sensitive variants of Random Forest and XGBoost (for example, `scale_pos_weight` in XGBoost) as an alternative or complement to SMOTE.
4. Conduct hyperparameter tuning (grid search or Bayesian optimisation) for the top two or three candidate models before final selection.
5. Reduce dimensionality of the one-hot encoded medication and diagnosis features — for example by grouping rare ICD-9 diagnosis codes into clinically meaningful categories — to improve model generalisation.

12. Conclusion

This project successfully applied the complete Data Analytics Lifecycle to a real-world clinical prediction problem: forecasting 30-day hospital readmission in diabetic patients. Starting from a clearly defined business problem and success metrics (Discovery), the project progressed through rigorous data cleaning and feature engineering (Data Preparation), exploratory analysis that surfaced a severe class imbalance (Model Planning), SMOTE-corrected training of four progressively complex classifiers (Model Building), a comprehensive multi-metric evaluation (Communicate Results), and a verified, deployable model artefact (Operationalization).

The most valuable outcome of this project is not simply the identification of a “best” model, but the demonstration that the choice of evaluation metric fundamentally changes which model should be deployed. XGBoost wins on ROC-AUC; Random Forest wins decisively on Recall — the metric the project itself identified, at the very start, as the one that matters most. Carrying that Discovery-phase decision through consistently to model selection, rather than defaulting to a generic leaderboard metric, is the single most important lesson this project offers for future iterations and for any team operationalising a similar clinical risk model.

Appendix A: Glossary of Key Terms

One-Hot Encoding (OHE)

Converts categorical text columns into binary 0/1 columns, one per category, without imposing an artificial numeric ordering. `drop_first=True` removes one column per group to avoid the dummy variable trap (perfect multicollinearity).

Train / Test Split

Dividing data into a training set (used to fit the model) and a held-out test set (used only for final evaluation). Stratified splitting preserves the original class ratio in both partitions — essential when classes are imbalanced.

SMOTE

Synthetic Minority Over-sampling Technique: generates new synthetic minority-class examples by interpolating between existing minority samples in feature space, applied only to training data to avoid leaking information into the test set.

Logistic Regression

A linear model that outputs the probability of class membership via the sigmoid function. Fast and interpretable, but assumes a linear decision boundary.

Decision Tree

A flowchart-like model that predicts by asking a sequence of yes/no questions about feature values. Highly interpretable but prone to overfitting without depth limits.

Random Forest

An ensemble of many decision trees (300 in this project), each trained on a random subset of data and features; predictions are made by majority vote. Averaging many trees reduces variance — a technique called bagging.

XGBoost

Extreme Gradient Boosting: an ensemble that builds trees sequentially, with each new tree correcting the errors of the previous ones. Often achieves state-of-the-art results on tabular data.

Precision

$TP / (TP + FP)$ — of all patients predicted to be readmitted, the proportion who actually were.

Recall (Sensitivity)

$TP / (TP + FN)$ — of all patients who were actually readmitted, the proportion the model correctly caught. The priority metric for this clinical use case.

F1 Score

The harmonic mean of Precision and Recall, balancing both into a single score.

ROC-AUC

Area under the Receiver Operating Characteristic curve; the probability that the model ranks a randomly chosen positive case above a randomly chosen negative case. 0.5 = random guessing, 1.0 = perfect separation.

Confusion Matrix

A 2×2 table of True Negatives, False Positives, False Negatives and True Positives. In this clinical context, False Negatives (missed readmissions) are considered more harmful than False Positives (unnecessary follow-up).

joblib

A Python library for efficiently saving and loading large objects, particularly trained scikit-learn and XGBoost models containing large NumPy arrays; used here to serialise the final model for deployment.

References

Dataset

Diabetes 130-US hospitals for years 1999–2008 (“diabetic_data.csv”). UCI Machine Learning Repository. The dataset represents ten years of clinical care records for diabetic patients across 130 US hospitals, including demographic, administrative, diagnostic, and medication information for approximately 100,000 encounters.

Source Materials

- Project notebook: `Copy_of_DA_healthcareReadmission_.ipynb` — contains the full data preparation, modelling, and evaluation pipeline summarised in this report.
- Project presentation: “Healthcare Patient Readmission Prediction Using Data Analytics Lifecycle” — source for the six-phase lifecycle framework structure used throughout this report.

Methodology Reference

CMS Hospital Readmissions Reduction Program (HRRP) — the regulatory context motivating the 30-day readmission window used as the prediction target in this project.

DATA ANALYTICS –P22CS601

ALTERNATE ASSESSMENT EVALUATION SHEET

Sl. No	Student Name	USN	Problem Identification and Objectives (1)	Data Collection and Dataset Description (1)	Data Preprocessig (Cleaning, Handling Missing Values, Transformation) (2)	Application of Data Analytics Techniques/ Algorithms (2)	Results, Visualization, and Interpretation (2)	Report Quality, Conclusion, and Presentation (2)	Problem – based Analysis (10)	Coursera (10)	Average (10)
1	Abhishek AR	4PS23CS001									
2	Ashwin S	4PS23CS014									
3	Aditya Chauhan	4PS23CS002									
4	Aditya manhas	4PS23CS003									
5	Preetham A	4PS24CS404									
6	Kushal S	4PS24CS401									
7	Harshith Gowda TS	4PS23CS056									
8	Hrudhay H	4PS23CS059									
9	Mohammed Adnan	4PS24CS402									
10	Chandhan C	4PS23CS032									

Faculty Signature with Date: